

Investigation Log: Zero Forcing Numbers of $P(n, k)$ for $k \geq 4$

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May 30 – September 28, 2026

Entry 1: How this problem was chosen

Date: May 30, 2026

This is the ninth research target attempted in this project. The previous eight were abandoned, and the reason matters: in every case the problem turned out to be already solved, and in every case I had inferred that it was open from the *absence* of a published result rather than from a positive statement.

- Four attempts on Turaev genus and signature bounds for positive braid closures — an area with an active professional research group.
- Periodicity of the octal game 0.45. I derived a real structural mechanism, then found the Guy–Smith periodicity theorem (1956), which settles it outright once one has computed past $n = 1018$. I had verified to 60,000.
- A “rare-index invariant” I derived for octal games, which is exactly Berlekamp’s Sparse Space Phenomenon, described in *Winning Ways* c. 1980.
- Persistent homology of linkage configuration spaces. Farber–Schütz (2007) give closed Betti-number formulas for planar polygon spaces, and Ba–Zhou (arXiv:2506.06781, 2025) prove the self-avoiding moduli spaces are diffeomorphic to Euclidean space — i.e. contractible, so persistent homology would correctly compute nothing at all.

The corrected procedure: before any investment, ask *what standard result would make this trivial*, and require a positive statement of openness. This target passes that test. Krishnan (arXiv:2607.19412, July 2026) closes with: “Determining $Z(P(n, k))$ for general k — the stabilization threshold and a rigorous lower bound for $k \geq 4$ — remains open.” That is an explicit assertion in a paper two months old, not an inference from silence.

Entry 5: A correction I had to make to my own claim

Date: June 18, 2026

After seeing $Z(P(n, 4)) = 10$ for $n = 18, \dots, 22$ — five consecutive values — I concluded the sequence had stabilized. The $k = 5$ row then showed $Z = 10$ at $n = 17, \dots, 21$, also five consecutive values, followed by 11 at $n = 22$ and 12 at $n = 23$. Five repeats are not evidence of stabilization. The $k = 4$ claim survives only because $2k + 2$ is a *proven* ceiling there, so the sequence cannot climb — but it dropped at $n = 12$ and $n = 16$, so it could in principle drop again. This is the same warning Krishnan gives about extrapolating from finite computation, arriving on my own data within an hour of my reading it.

Entry 9: A prediction, recorded before the test

Date: July 7, 2026

The thresholds at which $Z(P(n, k))$ first attains $2k + 2$ are 10, 13, 18, 23 for $k = 2, 3, 4, 5$. For $k \geq 3$ these are exactly $5k - 2$; $k = 2$ is the exception. The formula predicts $n_0(6) = 28$.

At the time of writing the $k = 6$ computation has reached $Z(P(27, 6)) = 13$, which is consistent (the ceiling 14 is not yet attained) but does not test the prediction. $P(28, 6)$ is running. This is recorded now, before the answer is known, so that a confirmation counts as a test rather than a fit — the same discipline that Entry 5 failed to apply.

Entry 10: The prediction was tested, and the conjecture died

Date: July 12, 2026

Entry 9 recorded, before the computation finished, that $n_0(6) = 5 \cdot 6 - 2 = 28$ and hence $Z(P(28, 6)) = 14$. The run took just over fourteen hours.

$Z(P(28, 6)) = 14$, with witness $\{u_0, \dots, u_{13}\}$. The narrow prediction was correct: 28 is the least n at which $k = 6$ attains its ceiling, exactly as $5k - 2$ says.

$Z(P(29, 6)) = 12$. And this refutes the conjecture. It was stated as “ $Z(P(n, k)) = 2k + 2$ for all $n \geq n_0(k)$ ” — a *stabilization* claim — and $29 > 28$ with $Z = 12 \neq 14$. So the prediction and the conjecture came apart: the thing I could check quickly was right, and the thing it was evidence for was wrong.

The cause is $\gcd(n, k)$, which had been visible in the $k = 6$ data for some time and which I twice set aside because $k = 4$ and $k = 5$ looked uniform. Sorting by $d = \gcd(n, 6)$, each class climbs on its own schedule: $d = 2$ reaches 14 at $n = 28$ while the coprime class has only reached 12 by $n = 29$. At $k = 4$ every class had already converged inside the computed range, so no separation was visible; at $k = 6$ the range finally became wide enough to see it. The apparent thresholds 10, 13, 18, 23 are where the slowest class happened to arrive, not values of a formula.

Two things worth recording about the process rather than the result. First, the prediction was useful precisely because it was written down in advance — had I fitted $5k - 2$ after seeing $Z(P(28, 6)) = 14$, I would have called it a confirmation and never computed $n = 29$. Second, this is the third time in two days that a pattern read off a short run has failed on the next data point (five repeats at $k = 5$; “ Z tracks n ” for I-graphs; now $5k - 2$). The consistent lesson is that in this family, sequences look stable well before they are.

Entry 11: A second conjecture, refuted by data I already had

Date: July 17, 2026

From 20 computed I-graphs, all satisfying $Z \leq 2(d_1 + d_2)$ with $d_1 = \gcd(n, j)$, $d_2 = \gcd(n, k)$, I wrote a conjecture into the paper asserting that bound in general. It is false, and the counterexamples were already in the validation table.

Every one of those 20 graphs had $d_1, d_2 > 1$ — that was the selection criterion for “genuinely new” I-graphs. Setting $j = 1$ gives $d_1 = 1$, and with $\gcd(n, k) = 1$ as well the bound reads $Z \leq 4$, while $Z(P(n, k)) = 2k + 2$ grows without bound. $Z(P(11, 3)) = 7$ and $Z(P(17, 4)) = 9$ refute it immediately. I had computed both values days earlier.

The error was generalising from a sample selected by a condition, to a statement about everything, without testing the boundary the condition excluded. The previous conjecture ($n_0 = 5k - 2$) at least took a fourteen-hour computation to kill; this one needed a table lookup.

What survives is narrower and still useful. The observation holds as stated for $d_1, d_2 > 1$, and a separate computation supports the underlying intuition: the minimum *bootstrapping* set for $I(18, 3, 8)$ — the object a uniform-in- n proof actually needs — has size $10 = 2(d_1 + d_2)$ against $2(j + k) = 22$, and decomposes as three consecutive vertices in each of the three outer cycles plus one inner vertex. Three consecutive vertices in an outer cycle is precisely what lets the middle one force its spoke. So the route to beating the $Z(I(n, j, k)) \leq 2(j + k)$ bound is a proof adapted to the cycle decomposition rather than to consecutive indices — but the target it proves must retain some dependence on j and k , not on d_1, d_2 alone.

Entry 14: A claim of mine, refuted — and the refutation is the result

Date: July 31, 2026

Date: September 26, 2026 **Command:** `python verification/d1_ceiling.py`

Entry 12 asserted that the period-one construction “never exceeds 6 for every k ”, because the symbol $F(s) = (s + a)L_k(s) + \alpha s + \beta$ has only three free parameters. I wrote a script to check that assertion exhaustively over all triples of grid roots, expecting it to confirm the claim. It printed 12.

The premise was right and the inference was wrong. At a root the condition is linear, so three roots determine (a, α, β) and no fourth can be imposed — but prescribing three roots *determines the other* $k - 2$, and those can land on the grid $\Gamma_n = \{2\cos(2\pi m/n)\}$ by themselves. They do so exactly when arithmetic makes them: a symbol with rational coefficients has Galois-closed roots, and the Galois orbit of $2\cos(2\pi m/n)$ stays inside Γ_n . One rational F places all $k + 1$ roots at once.

I had reasoned about how many roots I could *choose* and concluded something about how many roots could *be there*. The period-two machinery of Entry 12, built to escape a cap that was not real, is now redundant.

Entry 29: I deleted a correct citation because a summary told me to

Date: September 27, 2026

The reviewer retracted one of the eleven findings. Their W3 had said that arXiv:2607.19412 says nothing about $k \geq 4$, and that our sourcing of the $k \geq 4$ gap to that note was unsupported. I acted on it and deleted the sourcing. Then they pulled the PDF text and found the note’s final sentence:

“Determining $Z(P(n, k))$ for general k — the stabilization threshold and a rigorous lower bound for $k \geq 4$ — remains open.”

Our sourcing had been accurate and verbatim. I have restored it, having this time downloaded the PDF and read the sentence myself rather than taking anyone’s word for it, including the retraction’s.

What I want on the record is that this was my error too, not only theirs. When the report arrived I did check the claim before acting — but I checked it with a page fetch that returned the abstract plus a summarizer’s editorial remark that “the paper does not discuss $k \geq 4$.” I treated that sentence as evidence. It was not evidence; it was a guess by a tool with no access to the body, and the body is where the sentence lives. I then deleted correct text on the strength of it. The whole exchange is two parties making the same mistake in opposite directions, and the fix in both cases was ten seconds of `pdftotext`.

The rule I am taking from this: a claim about what a source does or does not say is a claim about its text, so it needs the text. Not the abstract, not a landing page, not a summary of either. I had already written a version of this rule for my own numerical results — quote the residual, not the impression — and I did not apply it to the literature.

Two consequences beyond the restoration. First, the conjecture we prove is *Conjecture 5*, numbered, and the note is sharper about it than I had conveyed: it says the content is the lower bound $Z(P(n, 3)) \geq 8$ for all $n \geq 13$, verified only to $n \leq 20$, that proving it means excluding 7-vertex forcing sets *uniformly in n* , and that “we will not assert it from finite computation, which would reproduce the gap this note corrects.” A paper that explicitly declines to extrapolate, and names uniformity in n as the missing step, is naming precisely what tiling supplies. The paper now says so and quotes it.

Second, I have applied the same standard against myself and *downgraded* a prior-art claim of my own. I had written that Duong–Kroschel–Riddell–Vander Meulen–Van Tuyl build their circulant matrices from orthogonal Hankel blocks. I have not read that paper — the publisher refused automated access three times — and that characterisation came from a summary of its abstract. So it is not a finding. The paper now cites the work as the first place a reader should look for overlap, records that we could not obtain the full text, and declines to say what is or is not in it. That is a gap in our literature review, stated as such.

And the bookkeeping, fixed at the root. The certified tile census has been 54, then 55, then 56, and $L(4)$ has been 31, 30, 29 — six numbers, six corrections, and not one of them a change to the theory. Each time I fixed the instances and each time it drifted again, because a count typed into prose is stale the moment the next tile certifies. So the

numbers are no longer typed. `verify_all.py` now reads the certificates off disk and emits `zf_numbers.tex`, which the paper and the board both `\input`. The count is now a build product. It was right to be told this three times; it was wrong of me to keep fixing instances instead of the cause.

What the suite was not checking at all. Seventy-odd checks, and not one of them checked a value of Z . Every headline claim in this project is a statement about $Z(P(n, k))$, and those numbers came out of one C program that nothing else ever tested — and the $k = 4$ values at $n = 24, \dots, 28$ depend on a rotation-symmetry reduction inside it that was likewise never tested. That is a strange blind spot to have while writing a section on how carefully the matrix side is certified. There are now cross-checks against an unreduced enumeration and against a CP-SAT integer program that does not enumerate subsets at all, over every value the paper prints, with no disagreement anywhere.

$k = 5$, by a route I had not noticed. All 21 identity tiles at $\ell = 54, \dots, 72, 74, 75$ certified, exactly. That does *not* give $[54, 108)$ — $\ell = 73$ and $76 \leq \ell \leq 107$ are still missing — so the tiling theorem does not apply as stated, and for a while I recorded $k = 5$ as still open. But $[L, 2L)$ is a sufficient condition dressed as the hypothesis; what the construction needs is only that n be a *sum* of certified lengths. The semigroup generated by those 21 lengths contains every integer $n \geq 162$. So $Z(P(n, 5)) = M(P(n, 5)) = 12$ for every $n \geq 162$, a fourth value of k , from tiles I already had. I had been reading my own hypothesis as a requirement.

Entry 33: the ceiling is not a property of the cover, and how the search lied

The question. Entry 30 left the ceiling proved for every matrix on $P(n, k)$ but only for *equivariant* matrices on a general base, and the paper recorded the gap as open: is the ceiling a property of the cover, or only of equivariant matrices on it? Closing it would have turned a fact about one family into a theorem about cyclic covers, which is why I went at it.

Two theorems first. The two-vertex elimination *chains*: on a path base with a loop at each vertex, solve the row at one end for the next fibre and repeat, each loop widening the window by $2p_t$, giving order $2 \sum p_t$ — exactly the degree span. That is Theorem ??, with the two-vertex case as $m = 2$. Cycle bases fall to something cruder: every base vertex has degree 2, so the cover is a disjoint union of $\gcd(|W|, n)$ cycles, and $M(C_N) = 2$ finishes it. Two different arguments, two different reasons.

Then the answer, which is no. The conjecture is false. A base vertex of degree 1 lifts to pendant vertices of the cover; the diagonal is free in $\mathcal{S}(G)$; so zero the pendant diagonals. The row at a pendant becomes $x_u = 0$ for its neighbour, the shared fibre dies, and what is left is one relation among $2n$ pendant coordinates. On the star-with-a-loop base, null $A = n$ against $D = 2$: the gap is not marginal, it grows without bound. Demanding minimum degree 2 does not save it either — $K_{2,3}$ gives 6 against $D = 4$. What survives is regularity, which costs nothing, since every cover this paper actually studies is cubic.

What I want to record is how I nearly missed it. I had run an adversarial search over ten connected bases and found no counterexample, and I wrote that into the paper as evidence *for* the conjecture. The star base was one of the ten. Three weaknesses compounded: Nelder–Mead in fifty dimensions, a penalty floor holding $|w| \geq 0.3$ — which excludes part of the admissible set rather than discouraging it — and an objective that summed the d smallest squared singular values, so one value at 10^{-10} hid two at 10^{-6} . Every one of those is a way of asking a weaker question than the one I thought I was asking.

What found it was not a better search. A gradient method flagged the base, and then the counterexample was written down *by hand* from the structure — pendants, free diagonal, zero — and verified by exact rank over $\mathbb{Z}$. The lesson is the one already on my own board, in the keybox about interpreting a search failure, and I walked straight past it: a failed search measures effort spent, not impossibility. The failed-search paragraph stays in the paper, relabelled. A search that does not find something is evidence about the search.

Three more of the same shape, in one day. An optimiser reported a strict gap $M(P(10, 2)) < Z$ that does not exist — two further random seeds found the nullity-6 basin. A `pgrep` wait matched its own command line and span forever; another failed to match at all, and I read a half-written file as a finished result. And a patch script died on a missing anchor inside an `&&` chain whose trailing `echo` printed “launched” anyway, so I announced a verification run that was not running, over a suite still asserting the conjecture I had just refuted. Different tools, one error: I took the success signal for the thing it was supposed to signal.

Entry 34: the obstruction, and what it is worth

Why regularity, finally. Entry 33 refuted the cover ceiling on an arbitrary base and left “regular” as a hypothesis fitted to the surviving cases, which is not a reason. The reason is one line of counting. Let S be independent and zero the diagonals on S : each row at $v \in S$ reduces to $a_{vv}x_v = 0$, which holds identically, and each row at a neighbour is one relation on $x|_S$, so vectors supported on S give $M(G) \geq |S| - |N(S)|$. On the star-with-a-loop cover that is $2n - n = n$ — exactly the nullity I had exhibited, so the bound is not merely compatible with the counterexample, it *is* the counterexample. And in a d -regular graph all $d|S|$ edges from an independent S land in $N(S)$, each vertex absorbing at most d , so $|N(S)| \geq |S|$ and the whole mechanism is unavailable. Now the hypothesis has a cause.

Then the thing I did not expect to find. I went looking for what the minimum rank community actually considers open, rather than guessing, and the Fallat–Hogben survey — which this paper already cites — asks for sufficient conditions giving $M(G) < Z(G)$ and for a parameter Y with $M \leq Y < Z$ explaining the gap. The degree span is a candidate Y , and I had been treating it as an internal bookkeeping device.

So I looked for a cover where Z exceeds it. On the K_4 base, $D = 6$ for every n while $Z = n + 2$, computed by CP-SAT with optimality proved at $n = 4, \dots, 8$. The certificate at the span came from the project’s own Galois trick pointed somewhere new: $\det M(\zeta)$ is palindromic, hence a cubic in $x = \zeta + \zeta^{-1}$, and the minimal polynomial of $2\cos\frac{2\pi}{7}$ is also a cubic, so matching them forces every coefficient equal and pins the weights to integers — edges 2, diagonals 4, 4, 5, 4, nullity exactly 6 by exact rank over $\mathbb{Z}$.

What is actually proved. The equivariant ceiling is a theorem, and the certificate attains it, so “equivariant maximum nullity is 6 while $Z = n + 2$ ” needs no conjecture: the gap is $n - 4$ and unbounded. Only the step from equivariant to all matrices is open, and that is the regular-base ceiling.

The honest size of it. I have not proved $M < Z$ on a cubic graph; I have proved that nothing rotation-invariant can close the gap, and conjectured the rest. The distinction is the whole difference between a result and a claim, and the board and the paper both state it that way. What changed is that the project now touches a question somebody else asked, which is not something I could say of it a week ago.

Entry 35: the gap family was the counterexample

1 October 2026. An outside read of the project, written from a summary and without the paper, pointed at the voltages of the K_4 base: the three edges of voltage 0 are $0-1$, $0-3$, $1-3$ — a triangle. Each fibre index carries a triangle, and the only edges leaving it go to the fibre over vertex 2. Make every triangle block rank one, $uu^\top$ with u entrywise nonzero: the pattern is respected, the diagonal is free, the rows at the $3n$ triangle vertices have rank at most $n + n$, and the rows over vertex 2 number n . So $\text{rank } A \leq 3n$ and $\text{null } A \geq n$. Exact rank over $\mathbb{Q}$ gives nullity exactly n for $n = 4, \dots, 10$ (verification/k4_rank_one.py).

What that kills. The conjecture $M(B^n) = 6$, and with it the cubic family with $Z - M \rightarrow \infty$ that Entry 34 called the strongest result in the project. $M \geq n$ and $Z = n + 2$, so the gap is at most 2. It kills the regular-base ceiling, since K_4 is cubic. And it kills a

sentence I had called unconditional, that the *equivariant* maximum nullity is 6: the rank-one matrix is equivariant and $\det M(\zeta) \equiv 0$ for it, so every block is singular. `thm:cover` needed the hypothesis $\det M(\zeta) \not\equiv 0$ and did not state it.

Why both attacks missed it. The diagonal-zeroing attack varies diagonals only; the gradient search starts from generic weights. The counterexample lives where three of the four fibre blocks are singular, which neither reaches. Entry 33 said a failed search is evidence about the search. This is the second conjecture the same lesson has cost.

What replaces regularity. The top coefficient of $\det M(\zeta)$, computed symbolically on every base in the paper. Where the ceiling is proved it is a monomial in edge weights alone: $w_1 w_k$ on the two-loop base, the product of the loop weights on a path with loops, the product of all edge weights on a cycle. On each of the three bases where the ceiling fails it contains a free diagonal: $-w_{12} w_{23} (w_{01} w_{03} - d_0 w_{13})$ on K_4 . That is the hypothesis every proof actually used, and it is now `conj:leading`. Regularity was a hypothesis fitted to the cases that survived two weak attacks; the general obstruction is $M(G) \geq |V| - \text{mr}(G[S]) - 2|V \setminus S|$ for any S , of which the independent-set bound of Entry 34 is the case $\text{mr}(G[S]) = 0$, and regularity does nothing to $\text{mr}(G[S])$.

Also corrected today. The abstract, the board and the arXiv draft said $Z = M = 2k + 2$ “for all n at $k = 2, 3, 4$ ”. Krishnan’s table has $Z(P(11, 3)) = Z(P(12, 3)) = 7$ and the paper itself has $Z(P(8, 2)) = 5$. The theorems carry the right thresholds; the summaries did not.

Found the same day. A pull request merged on 26 September 2026 in an automated proof repository claims a Lean-checked proof of Krishnan’s Conjecture 5 by an isoperimetric argument on forcing sets. It is not a paper and nobody has reviewed it, but it is dated, and nothing of mine is public.

References

- [1] S. Rashidi, N. Shajareh Poursalavati, M. Tavakkoli, *J. Algebra Combin. Discrete Struct. Appl.* 7(2) (2020), 183–193.
- [2] A. Krishnan, arXiv:2607.19412 (2026).

Entry 36: making the mathematics the spine

3 October 2026. [AI-DRAFTED — rewrite in your own words before this log is shown to anyone.] The question I set myself: how much of this project is mathematics and how much is computation, and can the line be moved. Three uses of the computer, three different statuses. The finite census and the small values of Z are finite case checks in exact arithmetic after a theorem has made the problem finite; nobody objects to that. The existence of the tiles is a computer-assisted existence proof: Krawczyk’s fixed-point theorem with its hypothesis verified in exact rational arithmetic, the four-colour kind of proof. The verification suite is bookkeeping and not mathematics at all, and I should stop presenting it as if it were evidence.

What moved. The closed form for $k \leq 9$ had been a proposition “verified case by case”. It is now a theorem: the negative set of D_k on $[-1, 1]$ is the end band(s) alone exactly for $k \in \{1, 2, 3, 4, 5, 7, 9\}$, by Sturm’s theorem (an integer computation on the coefficients, exact root isolation over $\mathbb{Q}$ in `verification/interior_bands.py`), and then only the grid point $j = 1$ — and for odd n and odd k only $j = (n - 1)/2$ — can enter a band. For $k = 2$ the whole argument fits on a page. From $k = 10$ on there is always an interior band, which is why the census is needed there.

A second theorem I should have seen earlier. The paper’s rotation-invariant certificates ($10 \mid n$ at $k = 3$; $60, 70, 90 \mid n$ at $k = 4$; $24 \mid n$ at $k = 5$) were constructions, with no claim that they were all of them. They can be classified. A rational symbol has a Galois-closed root set, a union of orbits $O_d = \{2 \cos(2\pi l/d) : \gcd(l, d) = 1\}$ of total size $k + 1$; the monic product of the minimal polynomials must equal $(s + a)(t_k(s) + d') - c_0$, which forces every middle coefficient; and realisability is $c_0 \neq 0$. That is a finite enumeration (`verification/period_one_classification.py`), and it gives an *if and only if*: a rotation-invariant matrix with rational symbol reaches nullity $2k + 2$ on $P(n, k)$ exactly when n is divisible by one of 7, 9, 15, 20 ($k = 2$); 10, 24, 42 ($k = 3$); 60, 70, 90 ($k = 4$); 24, 126 ($k = 5$); and never for $k = 6$, which recovers the $k = 6$ impossibility as a corollary. The classes 24 and 42 at $k = 3$ and 126 at $k = 5$ are new.

Still not pure. Tile existence. The paper already proves why forts, the fort LP and spectral bounds cannot give the all- n lower bound; a hand proof needs a new idea. Two candidates are written down in the handoff.